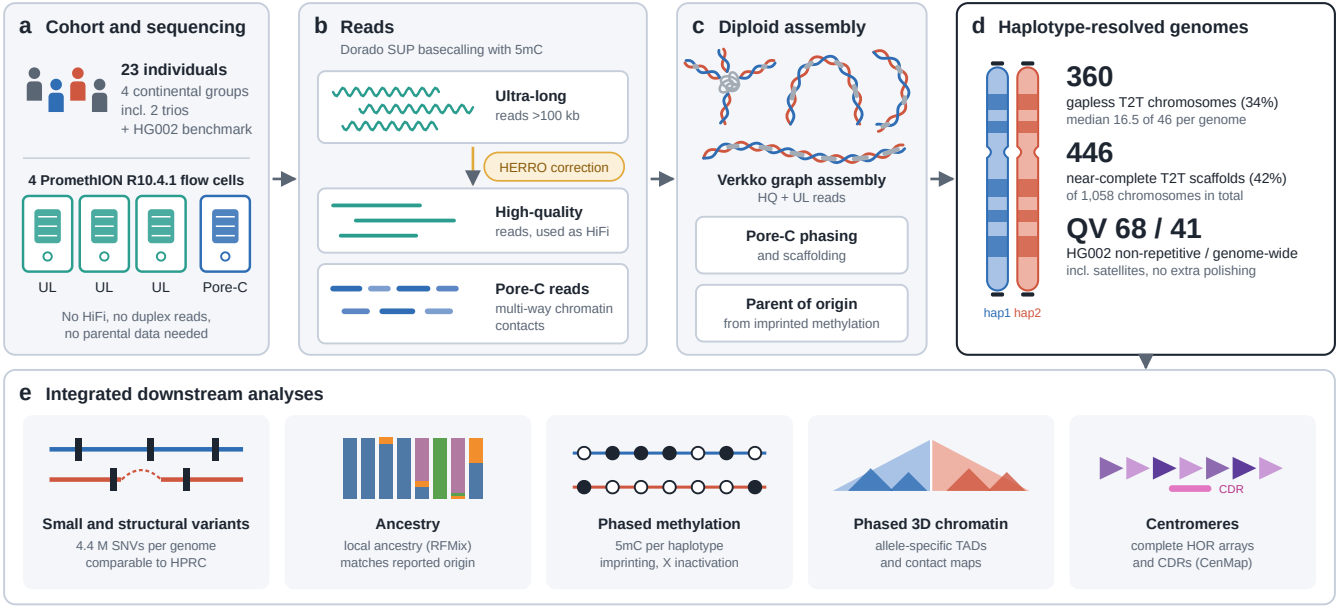

a Cohort and sequencing
23 individuals
4 continental groups
incl. 2 trios
+ HG002 benchmark
4 PromethION R10.4.1 flow cells
UL
UL
UL
Pore-C
No HiFi, no duplex reads,
no parental data needed
b Reads
Dorado SUP basecalling with 5mC
Ultra-long
reads >100 kb
HERRO correction
High-quality
reads, used as HiFi
Pore-C reads
multi-way chromatin
contacts
c Diploid assembly
Verkko graph assembly
HQ + UL reads
Pore-C phasing
and scaffolding
Parent of origin
from imprinted methylation
d Haplotype-resolved genomes
hap1 hap2
360
gapless T2T chromosomes (34%)
median 16.5 of 46 per genome
446
near-complete T2T scaffolds (42%)
of 1,058 chromosomes in total
QV 68 / 41
HG002 non-repetitive / genome-wide
incl. satellites, no extra polishing
e Integrated downstream analyses
Small and structural variants
4.4 M SNVs per genome
comparable to HPRC
Ancestry
local ancestry (RFMix)
matches reported origin
Phased methylation
5mC per haplotype
imprinting, X inactivation
Phased 3D chromatin
allele-specific TADs
and contact maps
CDR
Centromeres
complete HOR arrays
and CDRs (CenMap)